

# Advanced Topics

Probability Theory - Mathematics

## 1. Overview

This session introduces advanced topics within Probability Theory. It sets out the definitions used, the situations in which the material applies, and how it connects to the surrounding weeks of the module.

## 2. Key ideas

The central ideas of advanced topics are developed from first principles and then applied to worked examples. Emphasis is placed on the conditions under which each result holds, since applying a result outside its hypotheses is the most frequent source of error.

## 3. In practice

Exercises for this week apply advanced topics to problems of the kind met in assessment. Working them alongside the notes, rather than afterwards, is the intended use of this material.